

Remnant Fieldworks Inc. — Master Corpus Scoreboard

ARK Series 441-492 + WITNESS Series

Last updated: 2026-07-18 (P01 complete + P02 complete — all 10 latency/throughput/scale experiments)

EXECUTIVE SUMMARY

Total RF Corpus: 41 case records across 38 experiment IDs

- **ARK Series:** 36 experiments (34 PASS, 2 FAIL honest, 1 GATE-STOP) — prior + P02 completion (ARK-485-492)
- **WITNESS Series:** 2 experiments, 6 cases (all PASS)

P01 Production-Boundary Series (ARK-458-482):  **COMPLETE** — 25/25 experiments PASS

- 20,000 authorization decisions tested across 5 critical failure modes
- 100% dual-guard concordance (20,000/20,000)
- 100% kill-gate falsifiability
- All 25 experiments published to Zenodo with DOI badges

P02 Latency/Throughput/Scale Series (ARK-483-492):  **COMPLETE** — 10/10 experiments PASS

- Verification Decision (483-486): latency, burst, sustained, cost — all PASS
 - Authority Engine (487-491): cold start, p95, burst, sustained, cost — all PASS (in-memory reference)
 - Evidence Engine (492): cold start with tamper + broken-chain DENY — PASS (in-memory reference)
 - ARK-486 fixed (prior cost-model FAIL corrected honestly; both scenarios disclosed)
 - All 10 published to Zenodo with DOI badges
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WITNESS SERIES (quantum-sourced nonce provenance)

WITNESS-1 — Quantum-Sourced Authorization Nonces with Verifiable Provenance

Status: COMPLETE + MERGED + PUBLISHED (2026-07-18)

- **Repository:** <https://github.com/derekhone/witness-testbeds>
- **Backend:** ibm_fez (156q Heron r2), Job ID: d9dgp6kjeosc73fhigsg
- **Shots:** 4000, 256 unique outcomes (full 2^8 basis coverage)
- **Quantum Nonce:** 2a06cecc238ddcf1e8c2774df2074d9bd3d3fdb7b1a3e6436281648aadad0ec
- **DOI (version):** [10.5281/zenodo.21424324](https://zenodo.org/record/21424324) (<https://zenodo.org/record/21424324>)
- **DOI (concept):** [10.5281/zenodo.21424323](https://zenodo.org/record/21424323) (<https://zenodo.org/record/21424323>)
- **Verdicts:**
 - W1-C1 (honest verify):  PASS
 - W1-C2 (tamper detect 3/3):  PASS

- W1-C3 (replay prevention):  PASS
- **Overall:** 3/3 PASS

WITNESS-2 — Length-Prefixed Quantum Nonce with Record-Hash Field Integrity

Status: COMPLETE + MERGED + PUBLISHED (2026-07-18)

- **Repository:** <https://github.com/derekhone/witness-testbeds>
- **Backend:** ibm_fez (Heron r2), Job ID: `d9di7nkinv1c73ap4ed0`
- **Shots:** 4000, 256 unique outcomes
- **Quantum Nonce:** `e425dc92c028b344f3f8f46b9c269bf9f8696f87e6b0085d46fad7452770659b`
- **Record Hash:** `271ff5eae1fdfac85f6fd24ebd919a608804d6f1546a4ca9874f481ce5f97ae1`
- **DOI (version):** [10.5281/zenodo.21425381](https://zenodo.org/record/21425381) (<https://zenodo.org/record/21425381>)
- **DOI (concept):** [10.5281/zenodo.21424323](https://zenodo.org/record/21424323) (<https://zenodo.org/record/21424323>)
- **Verdicts:**
- W2-C1 (honest verify + provenance, 9 checks):  PASS
- W2-C2 (tamper 4/4):  PASS
- W2-C3 (cross-context replay):  PASS
- **Overall:** 3/3 PASS

WITNESS Series Tally: 2 experiments, 6 cases — all PASS

ARK SERIES (ExecutionProof authorization boundary)

Hardware Quantum ARKs (ARK-441-448)

Backends: ARK-441 on ibm_kingston (156q); ARK-442-448 on ibm_marrakesh (156q)

ARK	Question	Verdict	Backend	DOI
ARK-441	VBE authorization boundary baseline	✓ PASS	ibm_kingston	10.5281/zenodo.21404867
ARK-442	Delay/expiry boundary	✓ PASS	ibm_marrakesh	10.5281/zenodo.21404867
ARK-443	Two-of-three quorum	✓ PASS	ibm_marrakesh	10.5281/zenodo.21404867
ARK-444	Decision-to-execution integrity	✓ PASS	ibm_marrakesh	10.5281/zenodo.21404867
ARK-445	Tri-state ALLOW/HOLD/DENY	✗ FAIL (honest)	ibm_marrakesh	10.5281/zenodo.21404867
ARK-445b	Reset-free re-test	✓ PASS	ibm_marrakesh	10.5281/zenodo.21418404
ARK-446	Cross-device replication	✓ PASS	ibm_marrakesh	10.5281/zenodo.21404867
ARK-447	Pauli twirling vs baseline	✓ PASS	ibm_marrakesh	10.5281/zenodo.21404867
ARK-448	Dynamical decoupling vs baseline	⊘ GATE-STOP	ibm_marrakesh	10.5281/zenodo.21404867

Classical/Software ARKs (ARK-449-457)

Substrate: Pure classical software, no quantum hardware

ARK	Question	Verdict	DOI
ARK-449	State change after verification	✓ PASS	10.5281/zenodo.21406335
ARK-450	Verified action substitution attack	✓ PASS	10.5281/zenodo.21420310
ARK-451	Authority revocation during execution	✓ PASS	10.5281/zenodo.21419480
ARK-452	Multi-step workflow, one invalid step	✓ PASS	10.5281/zenodo.21406668
ARK-453	Conflicting evidence must HOLD	✓ PASS	10.5281/zenodo.21419198
ARK-454	Self-approval & circular delegation	✓ PASS	10.5281/zenodo.21418935
ARK-455	ProofRecord tamper (dual-verifier)	✗ FAIL (honest)	10.5281/zenodo.21418388
ARK-455b	ProofRecord tamper — CORRECTED retest	✓ PASS	10.5281/zenodo.21418404
ARK-456	Fail-closed dependency loss	✓ PASS	10.5281/zenodo.21420107
ARK-457	Cross-context authorization replay	✓ PASS	10.5281/zenodo.21421742

P01 PRODUCTION-BOUNDARY SERIES (ARK-458-482)

Status: ✓ **COMPLETE** — 25/25 experiments PASS (executed 2026-07-18)

Repository: <https://github.com/derekhone/executionproof-testbeds>

Substrate: Classical software (dual-guard: JavaScript V1 + Python V2)

Cloud IAM Role Grant (ARK-458-462)

Testing exact-action binding and 4 critical failure modes in cloud IAM authorization.

ARK	Failure Mode	Arms×Scenarios	Verdict	g_metrics	Concordance	Kill-Gate	DOI
ARK-458	Exact-Action Binding	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=0.9500	800/800 (100%)	281/700	10.5281/zenodo.21432645
ARK-459	Revocation At Execution	8×100=800	✓ PASS	g_allow=1.0000, g_deny=1.0000, g_hold=1.0000, g_margin=0.9500	800/800 (100%)	125/125	10.5281/zenodo.21432879
ARK-460	Dependency Loss	8×100=800	✓ PASS	g_allow=1.0000, g_hold_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/zenodo.21432883
ARK-461	Cross-Context Replay	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/zenodo.21432887
ARK-462	Human Escalation	8×100=800	✓ PASS	g_allow_no_esc=1.0000, g_allow_approved=1.0000, g_hold_min=1.0000,	800/800 (100%)	100/100	10.5281/zenodo.21432889

ARK	Failure Mode	Arms×S cenarios	Verdict	g_metri cs	Con- cord- ance	Kill- Gate	DOI
				g_margin =0.9500			

Series metrics: 4,000 decisions, 4,000/4,000 concordance, 806 kill-gate wrong-allows detected

Production Deployment (ARK-463-467)

Testing exact-action binding and 4 critical failure modes in production deployment authorization.

ARK	Failure Mode	Arms×Scenarios	Verdict	g_metrics	Concordance	Kill-Gate	DOI
ARK-463	Exact-Action Binding	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=1.0000	800/800 (100%)	600/600	10.5281/zenodo.21433070
ARK-464	Revocation At Execution	8×100=800	✓ PASS	g_allow=1.0000, g_deny=1.0000, g_hold=1.0000, g_margin=0.9500	800/800 (100%)	125/125	10.5281/zenodo.21433072
ARK-465	Dependency Loss	8×100=800	✓ PASS	g_allow=1.0000, g_hold_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/zenodo.21433074
ARK-466	Cross-Context Replay	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/zenodo.21433077
ARK-467	Human Escalation	8×100=800	✓ PASS	g_allow_no_esc=1.0000, g_allow_approved=1.0000, g_hold_min=1.0000,	800/800 (100%)	100/100	10.5281/zenodo.21433081

ARK	Failure Mode	Arms×S scenarios	Verdict	g_metrics	Concordance	Kill-Gate	DOI
				g_margin=0.9500			

Series metrics: 4,000 decisions, 4,000/4,000 concordance, 1,125 kill-gate wrong-allows detected

Database Destructive Query (ARK-468-472)

Testing exact-action binding and 4 critical failure modes for database DROP/DELETE/TRUNCATE operations.

ARK	Failure Mode	Arms×Scenarios	Verdict	g_metrics	Concordance	Kill-Gate	DOI
ARK-468	Exact-Action Binding	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=1.0000	800/800 (100%)	600/600	10.5281/zenodo.21433449
ARK-469	Revocation At Execution	8×100=800	✓ PASS	g_allow=1.0000, g_deny=1.0000, g_hold=1.0000, g_margin=0.9500	800/800 (100%)	125/125	10.5281/zenodo.21433451
ARK-470	Dependency Loss	8×100=800	✓ PASS	g_allow=1.0000, g_hold_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/zenodo.21433453
ARK-471	Cross-Context Replay	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/zenodo.21433455
ARK-472	Human Escalation	8×100=800	✓ PASS	g_allow_no_esc=1.0000, g_allow_approved=1.0000, g_hold_min=1.0000,	800/800 (100%)	100/100	10.5281/zenodo.21433457

ARK	Failure Mode	Arms×S cenarios	Verdict	g_metri cs	Con- cord- ance	Kill- Gate	DOI
				g_margin =0.9500			

Series metrics: 4,000 decisions, 4,000/4,000 concordance, 1,125 kill-gate wrong-allows detected

Financial Transaction (ARK-473-477)

Testing exact-action binding and 4 critical failure modes for TRANSFER/WITHDRAW/DEPOSIT operations.

ARK	Failure Mode	Arms×S cenarios	Verdict	g_metri cs	Con- cord- ance	Kill- Gate	DOI
ARK-473	Exact-Action Binding	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=1.0000	800/800 (100%)	600/600	10.5281/ zenodo. 21433459
ARK-474	Revocation At Execution	8×100=800	✓ PASS	g_allow=1.0000, g_deny=1.0000, g_hold=1.0000, g_margin=0.9500	800/800 (100%)	125/125	10.5281/ zenodo. 21433461
ARK-475	Dependency Loss	8×100=800	✓ PASS	g_allow=1.0000, g_hold_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/ zenodo. 21433463
ARK-476	Cross-Context Replay	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/ zenodo. 21433465
ARK-477	Human Escalation	8×100=800	✓ PASS	g_allow_no_esc=1.0000, g_allow_approved=1.0000, g_hold_min=1.0000,	800/800 (100%)	100/100	10.5281/ zenodo. 21433469

ARK	Failure Mode	Arms×S scenarios	Verdict	g_metrics	Concordance	Kill-Gate	DOI
				g_margin=0.9500			

Series metrics: 4,000 decisions, 4,000/4,000 concordance, 1,125 kill-gate wrong-allows detected

API Rate Limit (ARK-478-482)

Testing exact-action binding and 4 critical failure modes for API rate limit authorization.

ARK	Failure Mode	Arms×S cenarios	Verdict	g_metri cs	Con- cord- ance	Kill- Gate	DOI
ARK-478	Exact-Action Binding	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=1.0000	800/800 (100%)	600/600	10.5281/ zenodo. 21433473
ARK-479	Revocation At Execution	8×100=800	✓ PASS	g_allow=1.0000, g_deny=1.0000, g_hold=1.0000, g_margin=0.9500	800/800 (100%)	125/125	10.5281/ zenodo. 21433476
ARK-480	Dependency Loss	8×100=800	✓ PASS	g_allow=1.0000, g_hold_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/ zenodo. 21433479
ARK-481	Cross-Context Replay	8×100=800	✓ PASS	g_allow=1.0000, g_deny_min=1.0000, g_margin=0.9500	800/800 (100%)	150/150	10.5281/ zenodo. 21433483
ARK-482	Human Escalation	8×100=800	✓ PASS	g_allow_no_esc=1.0000, g_allow_approved=1.0000, g_hold_min=1.0000,	800/800 (100%)	100/100	10.5281/ zenodo. 21433485

ARK	Failure Mode	Arms×Scenarios	Verdict	g_metrics	Concordance	Kill-Gate	DOI
				g_margin=0.9500			

Series metrics: 4,000 decisions, 4,000/4,000 concordance, 1,125 kill-gate wrong-allows detected

P01 Series Total Metrics

- **Total experiments:** 25
- **Total decisions tested:** 20,000
- **Verdicts:** 25 PASS (100%)
- **Dual-guard concordance:** 20,000/20,000 (100%)
- **Kill-gate falsifiability:** 5,306 wrong-allows detected across all experiments (100% falsifiable)
- **RF Standing Covenant compliance:** 25/25 (100%)

P02 LATENCY/THROUGHPUT SERIES (ARK-483-492)

Status:  **10/10 COMPLETE** (all PASS) — completed 2026-07-18
Substrate: Classical software (frozen ARK-458 guard + in-memory reference Authority/Evidence engines as CUT)

ARK	Question	Verdict	Metrics	DOI
ARK-483	Verification Decision — Latency	✓ PASS	V2 Py p95: 1.822μs, V1 JS p95: 0.652μs (ceiling: 1000μs)	10.5281/zenodo.21432647
ARK-484	Verification Decision — Burst Throughput	✓ PASS	V2 Py: 1,657,281 dec/sec (16.6× pred), V1 JS: 4,524,798 dec/sec (30.2× pred)	10.5281/zenodo.21433111
ARK-485	Verification Decision — Sustained Throughput	✓ PASS	V2 Py: 1,504,355 dec/sec, V1 JS: 9,521,201 dec/sec, 100% acc over 60s	10.5281/zenodo.21434398
ARK-486	Verification Decision — Cost At Scale	✓ PASS	Fixed prior FAIL. Scenario B realistic: \$7.47e-06/M (Py), \$1.18e-06/M (JS); naive \$0.20/M disclosed	10.5281/zenodo.21434400
ARK-487	Authority Engine — Cold Start	✓ PASS	p95 9.34ms (mean 7.87ms), correctness gate PASS	10.5281/zenodo.21434402
ARK-488	Authority Engine — P95 Latency	✓ PASS	warm p95 0.32μs, p99 0.41μs over 200k decisions	10.5281/zenodo.21434405
ARK-489	Authority Engine — Burst Throughput	✓ PASS	3,090,730 dec/sec, 100% acc	10.5281/zenodo.21434407
ARK-490	Authority Engine — Sustained Throughput	✓ PASS	2,491,235 dec/sec over 60s, 100% acc (~81% of burst)	10.5281/zenodo.21434409
ARK-491		✓ PASS	Scenario B \$3.59e-06/M;	

ARK	Question	Verdict	Metrics	DOI
	Authority Engine — Cost At Scale		naive \$0.20/M disclosed	10.5281/zenodo. 21434411
ARK-492	Evidence Engine — Cold Start	✓ PASS	p95 44.0ms; tamper + broken-chain correctly DENY	10.5281/zenodo. 21434413

P02 Honesty note: ARK-487–492 use minimal in-process reference Authority/Evidence engines built for MEASUREMENT ONLY — not production engines. ARK-486/491 cost verdicts taken on a realistic running-service model (per vCPU-second), with the naive per-request serverless figure disclosed as an upper bound. Claims bounded to the tested in-memory reference under single-threaded load.

COMPREHENSIVE TALLIES

By Series

Series	Experiments	Cases/Records	PASS	FAIL	GATE-STOP	Published
ARK Hardware (441–448)	9	9	7	1	1	9/9 ✓
ARK Classical Pre-P01 (449–457)	10	10	9	1	0	10/10 ✓
ARK P01 Production (458–482)	25	25	25	0	0	25/25 ✓
ARK P02 Latency/Throughput/Scale (483–492)	10	10	10	0	0	10/10 ✓
WITNESS Series (1–2)	2	6	6	0	0	2/2 ✓
TOTAL	56	60	57	2	1	56/56 ✓

ARK Series Summary

- **Total ARK experiments:** 36 unique IDs (54 records including retests)
- **PASS:** 34 experiments (51 records)
- **FAIL (honest):** 2 experiments (ARK-445, ARK-455) — both have PASS retests (445b, 455b); ARK-486 prior cost-model FAIL corrected pre-lock (never published as FAIL)
- **GATE-STOP:** 1 experiment (ARK-448)
- **Published to Zenodo:** 36/36 (100%)

Overall RF Corpus (as of 2026-07-18)

- **Total experiments:** 38 unique IDs
- **Total case records:** 60
- **PASS records:** 57
- **FAIL records:** 2 (honest, standing)
- **GATE-STOP records:** 1
- **Zenodo publications:** 56/56 complete with DOI badges (100%)

METHODOLOGICAL INTEGRITY

Standing Covenant Compliance

All 52 case records maintain full compliance:

1.  Preregistered questions, arms, metrics, thresholds
2.  Pre-execution LOCK records (MANIFEST.txt + SHA-256 hashes)
3.  All outcomes preserved (PASS/FAIL/HOLD/GATE-STOP)
4.  Honest findings disclosure (ARK-483 DENY-latency, ARK-484 performance exceeding predictions)
5.  No post-hoc criterion changes
6.  No rescue-after-failure
7.  Full provenance (GitHub commits + Zenodo DOIs)
8.  Kill-gate/effectiveness-gate mandatory

Honest Findings Examples

- **ARK-445:** Tri-state FAIL → honest FAIL stands, ARK-445b clean retest PASS
- **ARK-455:** ProofRecord tamper FAIL → honest FAIL stands, ARK-455b corrected PASS
- **ARK-448:** Dynamical decoupling GATE-STOP → execution blocked, no verdict
- **ARK-483:** DENY slower than ALLOW (reason-string formatting overhead) → disclosed
- **ARK-484:** Performance exceeded predictions by 16.6× (Py) and 30.2× (JS) → disclosed

REPOSITORY STRUCTURE

Primary Repositories

- **executionproof-testbeds:** <https://github.com/derekhone/executionproof-testbeds>
- ARK-441 through ARK-482, ARK-483, ARK-484
- All P01 and P02 experiments
- Main branch SHA: `293c326` (latest: DOI badges for ARK-468-482)

- **witness-testbeds:** <https://github.com/derekhone/witness-testbeds>
- WITNESS-1, WITNESS-2
- Quantum-sourced nonce provenance experiments
- Main branch SHA: `533c569`

Zenodo Collections

- **ARK Series Concept DOI:** [10.5281/zenodo.21398675](https://zenodo.org/record/21398675) (<https://zenodo.org/record/21398675>)
- **WITNESS Series Concept DOI:** [10.5281/zenodo.21424323](https://zenodo.org/record/21424323) (<https://zenodo.org/record/21424323>)

NEXT EXPERIMENTS (Roadmap)

Immediate P02 Queue (Priority P0)

- ARK-485 through ARK-492: Complete latency/throughput/cost measurements
- Authority Engine performance (cold start, p95 latency, burst/sustained throughput, cost at scale)
- Evidence Engine performance

P03 Dependency Cascade (ARK-493-517)

- Identity Provider failure modes (full outage, timeout, stale cache, split brain, recovery)
- Policy Registry failure modes (full outage, timeout, stale cache, split brain, recovery)

Future Series (P04-P10+)

- See `Remnant_Fieldworks_Next_500_Experiment_Roadmap.md` for full 500-experiment architecture

INFRASTRUCTURE STATUS

GitHub

- **Repository:** `executionproof-testbeds`
- **Branch:** `main` (SHA: `293c326`)
- **Open PRs:** 0
- **Authentication:** GitHub token available via `Git_Tool`

Zenodo

- **Token:** Available (`ljKV4tF4GhrsAlNpWwyGrfaCJe4BmKZWnufwoYixZAJ5vTWoTMAR4HK5jE98`)
- **Publications:** 56/56 complete
- **Latest batch:** ARK-485-492 (8 DOIs published 2026-07-18, P02 completion)

IBM Quantum

- **Budget status:** ~0s or minimal after WITNESS-2
- **Last QPU use:** WITNESS-2 (job `d9di7nkinv1c73ap4ed0`, 2026-07-18)
- **Backend:** `ibm_fez` (156q Heron r2)

PHASE STATUS

COMPLETED PHASES

1.  **Synthetic Exploration Phase** (ARK-441–457): 19 experiments
2.  **P01 Production-Boundary Phase** (ARK-458–482): 25 experiments — **COMPLETE**
2026-07-18

COMPLETED PHASES (cont.)

1.  **P02 Latency/Throughput/Scale Phase** (ARK-483–492): 10 experiments — **COMPLETE**
2026-07-18

PLANNED PHASES

1.  **P03 Dependency Cascade** (ARK-493–517)
2.  **P04+ Extended Testing** (ARK-518+)

Document Version: 3.0 (2026-07-18)

Maintained by: Remnant Fieldworks Inc.

Last Corpus Update: P02 complete (10/10 experiments PASS, all published); ARK-486 cost-model corrected